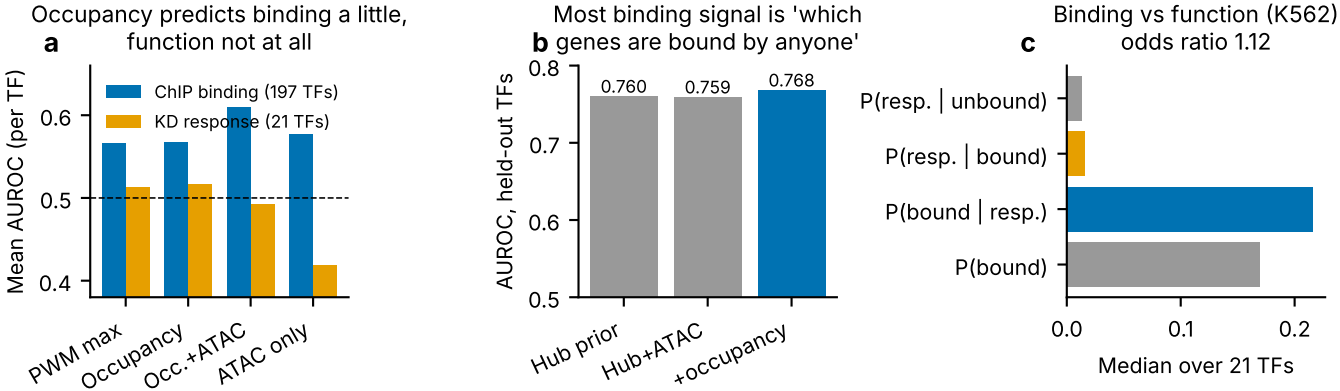
Occupancy predicts binding a little, function not at all
a
Mean AUROC (per TF)
ChIP binding (197 TFs)
KD response (21 TFs)
0.6
0.5
0.4
PWM max
Occupancy
Occ.+ATAC
ATAC only
Most binding signal is 'which genes are bound by anyone'
b
AUROC, held-out TFs
0.8
0.7
0.6
0.5
0.760
0.759
0.768
Hub prior
Hub+ATAC
+occupancy
Binding vs function (K562)
odds ratio 1.12
c
P(resp. | unbound)
P(resp. | bound)
P(bound | resp.)
P(bound)
0.0
0.1
0.2
Median over 21 TFs